UNIT CONVERTER TOOL

1. Project Overview

The Unit Converter Tool is a Python-based application developed to perform conversions between different units of measurement. The project provides a simple menu-driven interface where users can convert values related to length, weight, and temperature.

The main objective of this project is to understand the fundamentals of Python programming, including functions, conditional statements, loops, and user input handling. The application is designed to be beginner-friendly, efficient, and easy to use.

2. Intern Details

Field	Details
Intern ID	CITS1856
Full Name	Somya Singh
Project Name	Unit Converter Tool
Duration	4 Weeks

3. Project Scope

- Developing a console-based Python application
- Performing unit conversions accurately
- Creating a user-friendly menu-driven system
- Implementing multiple conversion categories
- Improving understanding of Python programming concepts

4. Features

- Length Conversion
- Weight Conversion
- Temperature Conversion
- Menu-driven interface
- User input validation
- Easy navigation
- Reusable functions

5. Technologies Used

- Python
- VS Code
- Command Line Interface

6. How to Run the Project

Step 1: Install Python
Step 2: Save the file as unit_converter.py
Step 3: Open Terminal or Command Prompt
Step 4: Run the program using: python unit_converter.py
Step 5: Choose conversion options from the menu

7. Learning Outcomes

• Understanding Python syntax and structure
• Working with functions and modular programming
• Using loops and conditional statements
• Taking user input and displaying formatted output
• Building menu-driven applications
• Improving logical thinking and problem-solving skills
• Debugging and testing Python programs

8. Conclusion

The Unit Converter Tool successfully demonstrates the practical implementation of Python programming concepts. The project provides accurate unit conversions through a simple and interactive interface. It also serves as a strong foundation for developing more advanced Python applications in the future.